



# WAN — ASSIGNMENT

## Topic: Wide Area Network (WAN) and WAN Technologies

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### QUESTION 1

Create a table comparing at least three types of WAN connections (such as leased line, DSL, and MPLS) with columns for speed, cost, and typical use cases.



### ANSWER

A **Wide Area Network (WAN)** is a network that connects computers, offices, branches, and other networks across large geographical areas such as cities, states, or countries.



### Comparison of WAN Connections

|  WAN Connection |  Typical Speed |  Cost        |  Typical Use Cases |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Leased Line</b>                                                                                | 10 Mbps – 10 Gbps+                                                                               |  High       | Banks, large enterprises, dedicated office-to-office connectivity                                     |
| <b>DSL</b>                                                                                        | 1 – 100+ Mbps                                                                                    |  Low        | Homes, small businesses, small branch offices                                                         |
| <b>MPLS</b>                                                                                       | 10 Mbps – several Gbps                                                                           |  High       | Enterprise branches, data centers, corporate networks                                                 |
| <b>Fiber Broadband</b>                                                                            | 100 Mbps – 1 Gbps+                                                                               |  Low–Medium | Branch offices, cloud applications, Internet access                                                   |



### Brief Explanation

#### ◆ Leased Line:

A leased line is a dedicated WAN connection provided by a telecom company. It provides consistent bandwidth and is suitable for organizations that require reliable connectivity.

#### ◆ DSL:

DSL uses telephone-line infrastructure to provide Internet connectivity. It is generally inexpensive and suitable for homes and small businesses.

#### ◆ MPLS:

MPLS is a service-provider-managed WAN technology used to connect multiple enterprise locations. It can provide traffic prioritization and predictable connectivity.

◆ **Fiber Broadband:**

Fiber broadband uses optical fiber to provide high-speed connectivity. It is commonly used by businesses for Internet access and cloud applications.

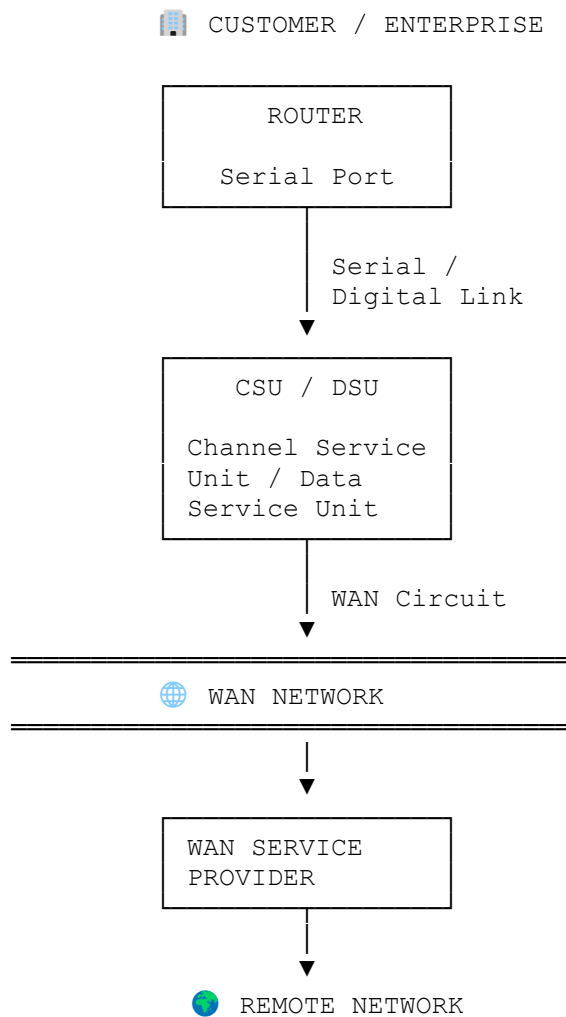
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## QUESTION 2

**Draw a labeled diagram (on paper or using any drawing tool) showing how a CSU/DSU connects a router to a WAN service provider, and explain the role of each device in 2–3 sentences.**

✓ **ANSWER**

 **CSU/DSU WAN Connection**



 **Router**

The **router** connects the organization's LAN to the WAN and forwards IP packets between local and remote networks. It uses routing information to select the appropriate path for network traffic.

### CSU/DSU

The **CSU/DSU (Channel Service Unit/Data Service Unit)** provides the interface between the customer's router and the digital WAN circuit. It performs functions such as signal conditioning, timing/clocking, and interfacing with the service provider's digital circuit.

### WAN Service Provider

The **WAN service provider** operates the telecommunications network that carries data between geographically separated locations. It provides the WAN circuit or service used by the customer's router.

 **Note:** In modern networks, CSU/DSU functionality may be integrated into networking equipment or supplied by the service provider, so a separate CSU/DSU device is not always visible.

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## QUESTION 3

List three real-world examples of popular apps or services (like IRCTC, WhatsApp, or Zomato) that rely on WAN connections, and briefly describe how WAN enables their functionality.

### ANSWER

#### 1. IRCTC

IRCTC provides online railway services to users throughout India.

##### **How WAN helps:**

WAN/Internet connectivity allows users from different cities to access IRCTC's online systems for **ticket booking, cancellation, train information, and other services**.

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#### 2. WhatsApp

WhatsApp is a messaging and communication service used worldwide.

##### **How WAN helps:**

WAN connectivity allows users in different cities and countries to exchange **messages, images, documents, voice calls, and video calls** through Internet-connected infrastructure.

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### 3. Zomato

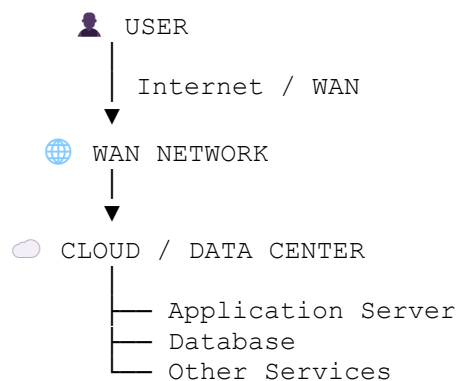
Zomato connects customers, restaurants, and delivery partners through its online platform.

#### How WAN helps:

WAN/Internet connectivity enables **online food ordering, restaurant information, payment processing, and delivery-status updates** between different users and backend systems.

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### How WAN Supports These Applications



WAN allows users and servers located in different geographical locations to communicate with each other.

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## QUESTION 4

Research online and find an image or datasheet of a physical WAN aggregator device used in enterprise networks. Paste the image or link, and write two key features of that device.

### ANSWER

#### Example: Cisco Catalyst 8300 Series Edge Platform

#### Official Cisco Product Page

[Cisco Catalyst 8300 Series Edge Platforms — Official Cisco Page](#)

#### Two Key Features

## 1 Enterprise WAN & Cloud Connectivity

The Cisco Catalyst 8300 is designed for enterprise branch and WAN connectivity. It can connect branch offices with **data centers, cloud services, and other enterprise locations**.

## 2 SD-WAN & Security Support

The platform supports **Cisco SD-WAN** capabilities and integrated security features. This helps organizations centrally manage WAN connections and apply networking and security policies.

### 💡 Why Is It Called a WAN Aggregator?

A **WAN aggregator** brings together WAN traffic and connections from different enterprise locations or network links. It can act as an important point for connecting **branch offices, data centers, cloud environments, and other networks**.

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## QUICK REVISION

|  Topic |  Key Point |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>WAN</b>                                                                               | Connects networks over large geographical areas                                              |
| <b>Leased Line</b>                                                                       | Dedicated and reliable WAN connection                                                        |
| <b>DSL</b>                                                                               | Low-cost WAN/Internet connection                                                             |
| <b>MPLS</b>                                                                              | Managed enterprise WAN technology                                                            |
| <b>CSU/DSU</b>                                                                           | Interfaces a router with a digital WAN circuit                                               |
| <b>WAN Provider</b>                                                                      | Provides the WAN connectivity/service                                                        |
| <b>WAN Aggregator</b>                                                                    | Combines/handles WAN connections and traffic                                                 |
| <b>Cisco Catalyst 8300</b>                                                               | Enterprise WAN/SD-WAN edge platform                                                          |

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## CONCLUSION

**WAN is an essential networking technology for connecting geographically separated networks, branches, data centers, cloud services, and users.** Different WAN technologies such as **Leased Line, DSL, MPLS, and Fiber Broadband** provide different combinations of speed, cost, and reliability.

Devices such as **routers and CSU/DSUs** provide connectivity between an organization and the WAN service provider, while modern enterprise platforms such as the **Cisco Catalyst 8300** support advanced WAN and SD-WAN requirements.

### ★ Key Takeaway

**“WAN connects people, branches, applications, and networks across geographical distances.”**

